Fiscal Year:	FY 2004	**Task Last Updated:**	FY 04/03/2006
PI Name:	Shapiro, Jay R M.D.		
Project Title:	Defining and Preventing Bone Loss: A Model for Microgravity		
Division Name:	Human Research		
Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Bone Loss Team		
Joint Agency Name:	National Institutes of Health/National Cancer Institute	**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Organization Type:	UNIVERSITY	**Phone:**	301-295-3600
Organization Name:	Uniformed Services University		
PI Address 1:	4301 Jones Bridge Road		
PI Address 2:			
PI Web Page:			
City:	Bethesda	**State:**	MD
Zip Code:	20814	**Congressional District:**	8
Comments:	This research is sponsored by NSBRI		
Project Type:	Ground	**Solicitation / Funding Source:**	NSBRI
Start Date:	02/01/2001	**End Date:**	09/30/2004
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NCC 9-58-BL.002.02		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	This project proposes to evaluate the spinal cord injured subjects, either tetraplegic or paraplegic, as a model for the bone loss that is observed during space flight. The project will employ, in a double blind, placebo-controlled, randomized mode, the effectiveness of zoledronate (Novartis) as a countermeasure to bone loss in these subjects who are non-weight bearing. Zoledronate is a potent new third generation bisphosphonate which can be administered intravenously over a 15 minute period, with a a prolonged duration of action, lasting up to one year after a single administration. This program will test its effectiveness in preventing bone loss in this model of microgravity.		
Rationale for HRP Directed Research:			

Research Impact/Earth Benefits:	Non-weightbearing is common to many populations: children with inherited, acquired and traumatic musculoskeletal disorders as well as adults with immobilization due to disease, injury or aging. Astronauts are a unique example of non-weightbearing associated with significant bone loss and increased risk of fracture. A countermeaure to bone loss is a NASA priority. The ability to lessen or prevent bone loss in this model of microgravity will benefit each of these populations, thus it has broad application to individuals on earth.
Task Progress:	No progress report this period.
Bibliography Type:	Description: (Last Updated: 02/27/2004)